

Farm Management System

Daily Progress - QR, Economics, Scan & Breeding

Date: 22 August 2026 | Scope: today's work only

Focus: QR tags, invested/earned, Scan hub popup, breeding history on animal page

1. What we did today

Added printable QR for breeding livestock and shed/pond batches, animal/batch economics (invested vs earned), an in-app Scan QR hub with camera + popup, and breeding history on the breeding-stock animal detail page.

1.1 Data model & migration

- Animal.breedingStock - flag breeding parents
- Expense / Revenue animalId + herdBatchId - attribute money to a target
- HealthRecord.cost - optional vaccine/treatment cost (NPR)
- Migration: 20260822120000_animal_qr_economics

1.2 Economics APIs

- GET /v1/animals/:id/economics and GET /v1/batches/:id/economics
- Invested = purchase (animals) + linked expenses + health costs
- Earned = linked revenue; Net = earned - invested
- Expense / revenue / health forms can link animal or batch (incl. deep links)

1.3 Printable QR + Scan hub

- QR on /animals/stock/:id and /batches/:id (Print QR)
- URL: /scan/a/:id (animal) or /scan/b/:id (batch)
- Nav module Scan QR -> /scan with live camera (html5-qrcode)
- Popup shows economics + quick actions (expense, health, revenue, breeding)
- Manual lookup by animal tag or batch name if camera unavailable
- en + ne copy for QR / scan strings

1.4 Breeding history on animal detail

- Breeding history card on /animals/stock/:id (was missing before)
- Shows mating type, mating/due dates, days left, status, offspring, father/Al
- GET /v1/breeding?motherId= filters records for that animal
- Add mating opens /breeding?animalId= with mother preselected
- DTO includes motherTag + daysRemaining for list/UI

2. How it works

2.1 Print -> scan -> popup

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Print QR on animal / batch detail
  |
  v
Encodes /scan/a/:id or /scan/b/:id
  |
  +--> Phone camera deep link --> popup
  +--> In-app Scan QR camera --> popup
  +--> Manual tag / batch lookup --> popup
    |
    v
Economics API + quick actions

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2.2 Animal page layout (today)

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/animals/stock/:id
- Hero + QR / Edit / Delete
- Invested / Earned / Net tiles
- Basic information (incl. Breeding stock Yes/No)
- Printable QR card
- Weight history
- Breeding history (NEW) + Add mating link

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2.3 Economics formula

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Animal: Invested = purchase + expenses + health.cost
        Earned   = revenue; Net = Earned - Invested
Batch:  Invested = expenses + health.cost (no purchase)
        Earned   = revenue; Net = Earned - Invested

```

3. Screens & APIs added/changed today

Route	Purpose
/scan	Scan hub - camera + lookup + popup
/scan/a/:id, /scan/b/:id	QR deep links -> same popup
/animals/stock/:id	QR, economics tiles, breeding history
/batches/:id	Shed/pond QR + invested/earned
/breeding?animalId=	Add mating prefilled for mother
/expenses, /health, /revenue	Optional animal / batch link

API endpoints

- GET /v1/animals/:id/economics
- GET /v1/batches/:id/economics
- GET /v1/breeding?motherId= (filter for animal page)
- POST expense/revenue/health accept animalId / herdBatchId; health cost optional

How to try

- Open breeding animal -> see economics + QR + Breeding history card
- Add mating from animal page if history is empty
- Open Scan QR, look up BUF001 (or scan a printout) -> popup totals
- Link an expense to the animal; refresh economics / scan again
- Camera needs HTTPS or localhost; use tag lookup if camera is blocked

This PDF covers only work completed on 22 August 2026. Earlier features (batches, money modules, etc.) are not restated here.